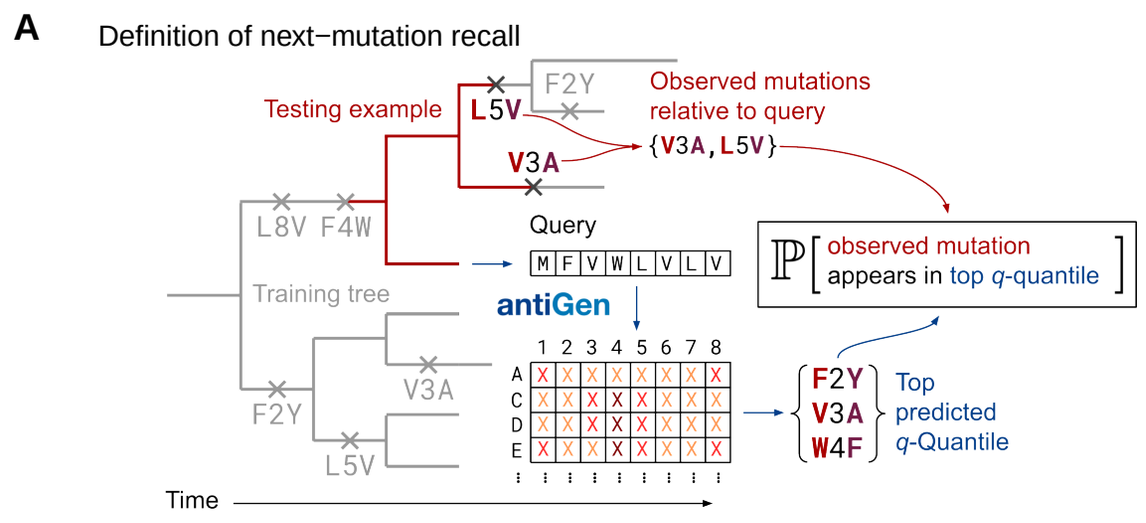
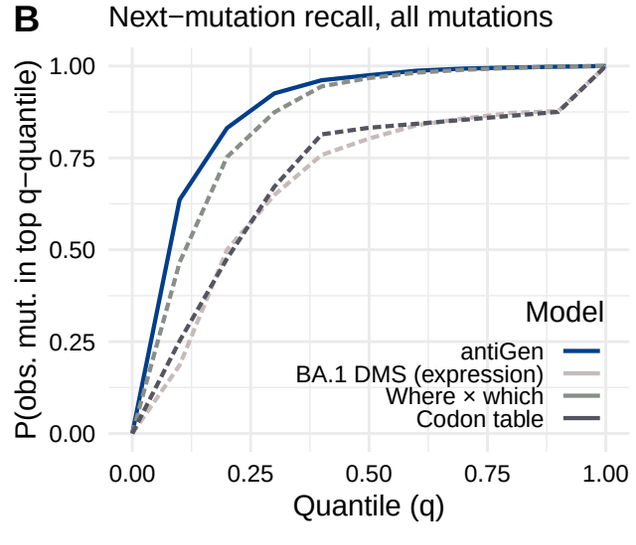
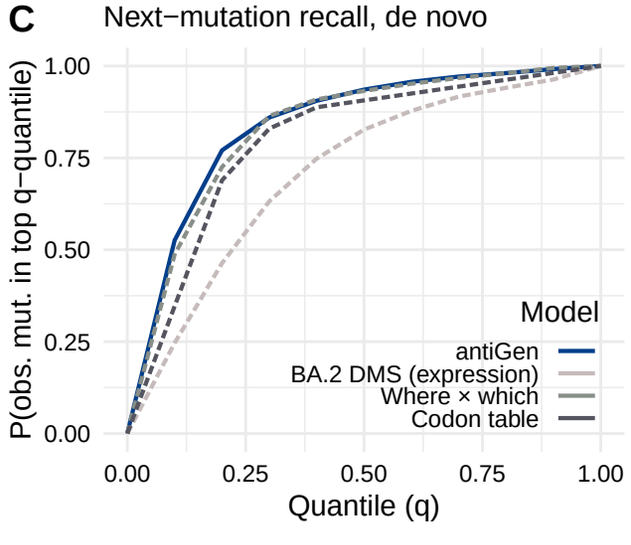
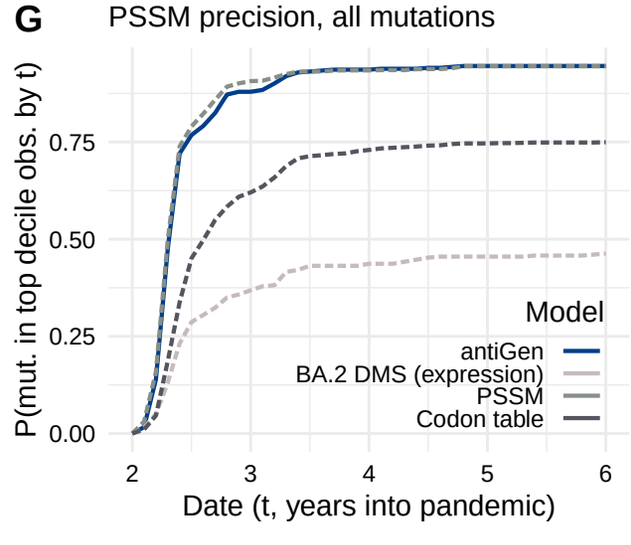
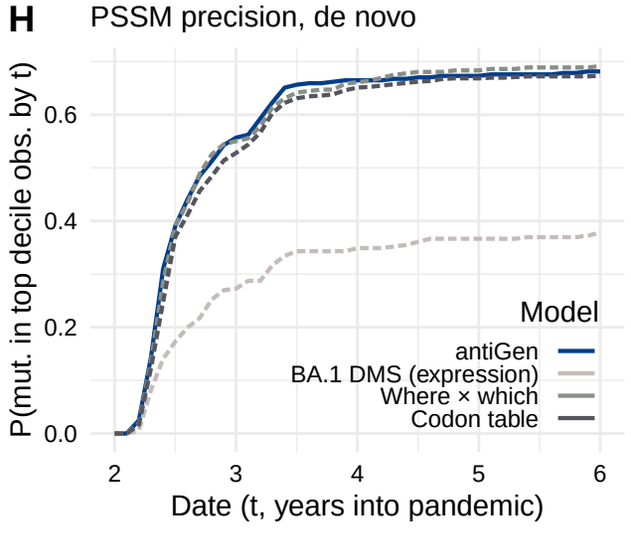

A
Definition of next−mutation recall
Testing example
L5V
F2Y
V3A
Observed mutations relative to query
{V3A, L5V}
L8V F4W
Query
M F V W L V L V
Training tree
antiGen
1 2 3 4 5 6 7 8
A C D E
ℙ[observed mutation appears in top q-quantile]
F2Y
V3A
W4F
Top predicted q-Quantile
Time

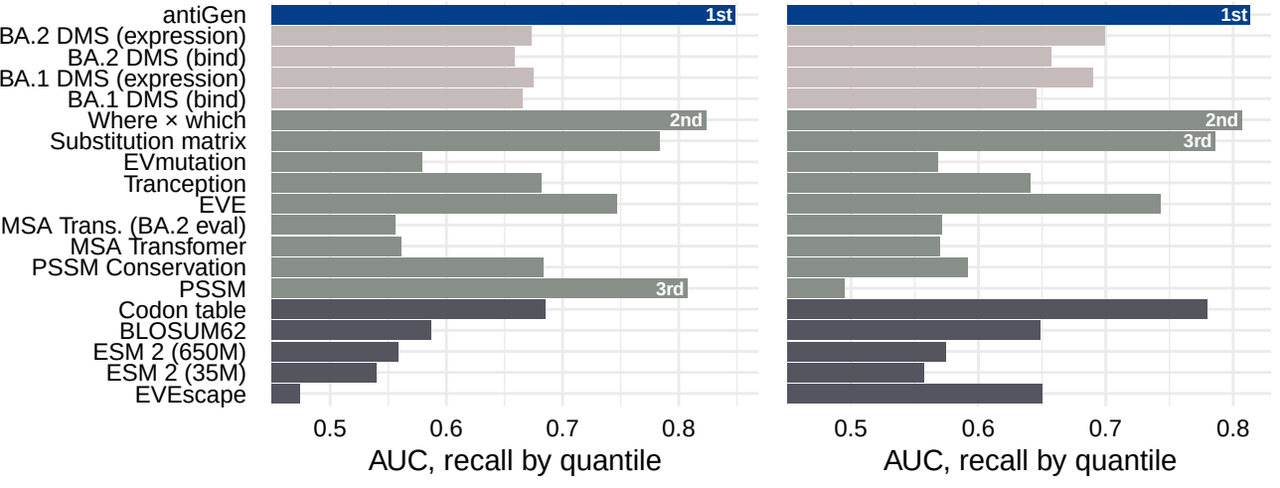

B
Next−mutation recall, all mutations
P(obs. mut. in top q−quantile)
Quantile (q)
Model
antiGen
BA.1 DMS (expression)
Where × which
Codon table


C
Next−mutation recall, de novo
P(obs. mut. in top q−quantile)
Quantile (q)
Model
antiGen
BA.2 DMS (expression)
Where × which
Codon table

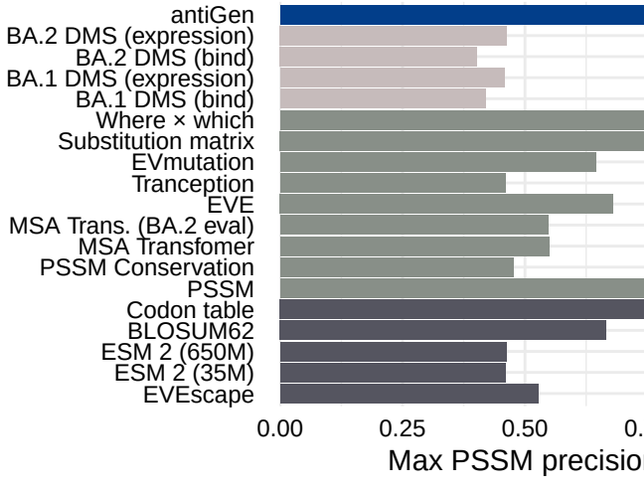
F
Definition of PSSM precision
Testing data downstream of query
F2Y
L5V
V3A
PSSM downstream of query
L8V F4W
Query (VOI)
M F V W L V L V
Training tree
antiGen
1 2 3 4 5 6 7 8
A C D E
ℙ[prediction in top decile appears at frequency ≥ M]
F2Y
W4F
Top predicted decile
Time

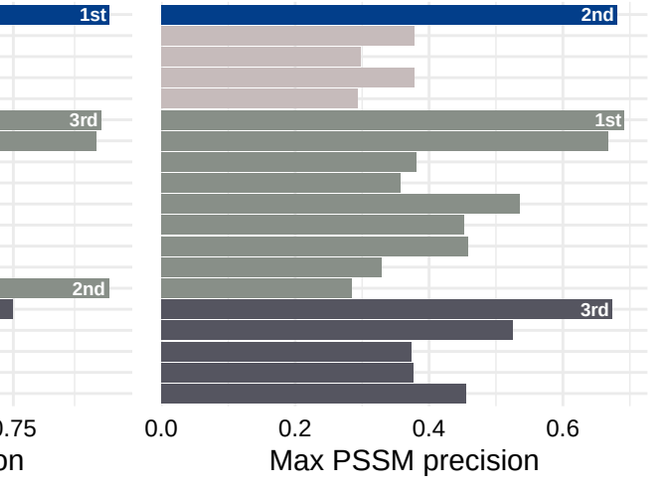

G
PSSM precision, all mutations
P(mut. in top decile obs. by t)
Date (t, years into pandemic)
Model
antiGen
BA.2 DMS (expression)
PSSM
Codon table

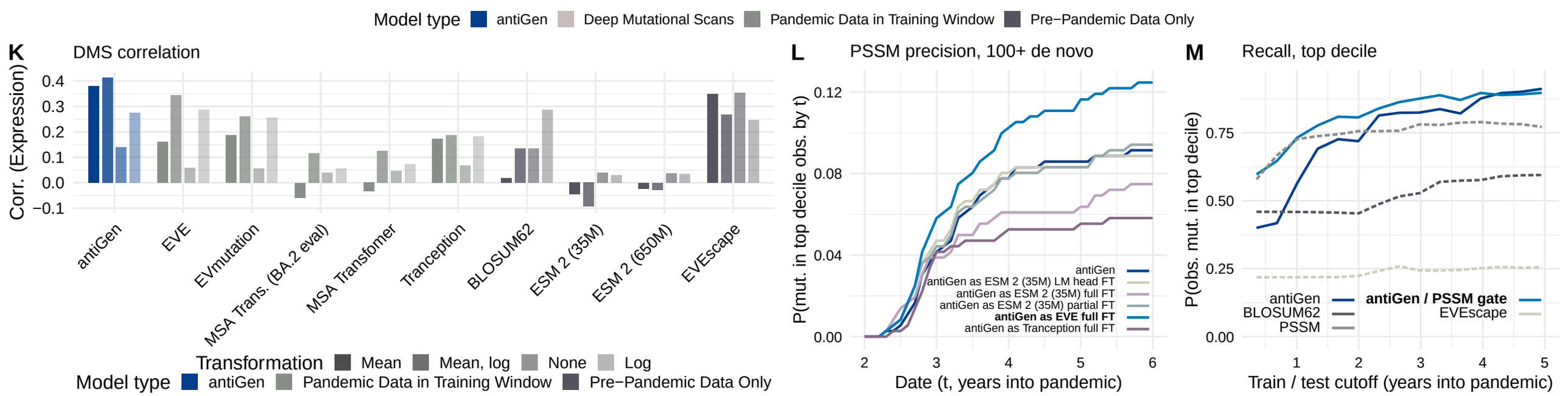

H
PSSM precision, de novo
P(mut. in top decile obs. by t)
Date (t, years into pandemic)
Model
antiGen
BA.1 DMS (expression)
Where × which
Codon table


D
Recall AUC, all mutations
antiGen
BA.2 DMS (expression)
BA.2 DMS (bind)
BA.1 DMS (expression)
BA.1 DMS (bind)
Where × which
Substitution matrix
EVmutation
Tranception
EVE
MSA Trans. (BA.2 eval)
MSA Transfomer
PSSM Conservation
PSSM
Codon table
BLOSUM62
ESM 2 (650M)
ESM 2 (35M)
EVEscape
1st
2nd
3rd
AUC, recall by quantile

E
Recall AUC, de novo
1st
2nd
3rd
AUC, recall by quantile


I
Max precision, all mutations
1st
2nd
3rd
Max PSSM precision


J
Max precision, de novo
1st
2nd
3rd
Max PSSM precision


Model type
antiGen
Deep Mutational Scans
Pandemic Data in Training Window
Pre−Pandemic Data Only

K
DMS correlation
Corr. (Expression)
antiGen
EVE
EVmutation
MSA Trans. (BA.2 eval)
MSA Transfomer
Tranception
BLOSUM62
ESM 2 (35M)
ESM 2 (650M)
EVEscape
Transformation
Mean
Mean, log
None
Log
Model type
antiGen
Pandemic Data in Training Window
Pre−Pandemic Data Only

L
PSSM precision, 100+ de novo
P(mut. in top decile obs. by t)
Date (t, years into pandemic)
antiGen
antiGen as ESM 2 (35M) LM head FT
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (35M) partial FT
antiGen as EVE full FT
antiGen as Tranception full FT

M
Recall, top decile
P(obs. mut. in top decile)
Train / test cutoff (years into pandemic)
antiGen
BLOSUM62
PSSM
antiGen / PSSM gate
EVEscape